

Bethel Hall

Hoboken, NJ | +1 908-880-4895 | bhall2@stevens.edu | bethelhall.tech | linkedin.com/in/bethelhall

EDUCATION

Stevens Institute of Technology

Ph.D. in Computer Science

Sep. 2022–Present

Hoboken, NJ

- Advisor: Prof. William Eiers; GPA: 3.92 / 4.00.
- Research Interests: AI for Software Engineering, Pretraining & Post-Training, Formal Verification, Neurosymbolic AI, Trustworthy AI Systems.
- Expected graduation: May 2028.

Addis Ababa University

B.Sc. in Software Engineering

June 2022

Addis Ababa, Ethiopia

RESEARCH EXPERIENCE

Stevens Institute of Technology

Graduate Research Assistant, Department of Computer Science

Sep. 2022–Present

Hoboken, NJ

VeriFin: Verification of LLM-Generated Financial Claims

Methods & Tools: Z3, SMT, XBRL, SEC EDGAR, Hybrid Retrieval, Cross-Encoder Reranking, LLMs

- Built a neurosymbolic verification framework that grounds LLM-generated financial claims in filed XBRL facts and independently derives authorized calculations before symbolic verification.
- Developed provenance-aware verification over financial concepts, reporting periods, units, and XBRL calculation linkbases, compiling claims into arithmetic constraints checked by Z3 with unsat-core feedback.
- Created XBRLFiling, a 600-question benchmark from 86 SEC 10-K filings across 28 companies; achieved zero false accepts on incorrect claims.

CloudFix: Automated Repair of Cloud Access Control Policies

- Built an automated repair system for AWS IAM policies combining SMT-based statement-level fault localization with targeted LLM-generated edits.
- Implemented symbolic policy validation by translating IAM semantics to SMT constraints and checking allow/deny request specifications with Z3, constraining repair to formally localized faults.
- Achieved 84.0% complete repair versus 48.2% for the LLM-only baseline across 282 real-world AWS policies with 10-request specifications.

VeriMed: Neurosymbolic Auditing of Software Requirements

- Built a neurosymbolic auditing system that samples independent formal encodings of natural-language safety requirements and detects ambiguity through bidirectional SMT equivalence checking.
- Designed a counterexample-guided repair loop that converts solver disagreements into targeted feedback for refining specifications and subsequent LLM reasoning.
- Improved SMT-verified question-answering accuracy from 55.4% to 98.5% across 65 hemodialysis safety questions.

PolicySummarizer: Verified Characterization of Access Control Policies

- Built a neurosymbolic policy-analysis system that converts cloud access control semantics into automata-derived regular expressions and uses an LLM to produce compact human-readable characterizations.
- Developed bounded verification using ABC model counting and language comparison to reject LLM-generated characterizations that diverge from the formally derived request language.
- Evaluated across 746 AWS, Azure, and GCP policies, achieving 0.93 mean Jaccard similarity for resource characterizations.

University of Michigan

Undergraduate Research Assistant

2021

Ann Arbor, MI

- Proposed and developed safety-aware neural network training method by incorporating formally specified constraints into the learning objective.

TECHNICAL SKILLS

LLMs & Post-Training: Hugging Face Transformers, vLLM, PyTorch; SFT, DPO, GRPO, Distillation, RAG, Agentic Systems.

Formal & Neurosymbolic Methods: Z3, SMT/SMT-LIB, LLM Autoformalization, Counterexample-Guided Repair, Finite-State Automata, Model Counting, Lean, LTL.

Systems & Domains: XBRL, SEC EDGAR, AWS IAM, Azure RBAC, GCP IAM, Quacky, ABC, Docker, AWS, Azure, GCP.

Programming: Python, SQL, Java, R, $\LaTeX$.

SELECTED PUBLICATIONS

1. **Bethel Hall et al.** “CloudFix: Automated Policy Repair for Cloud Access Control Policies Using Large Language Models.” SANER, 2026.
2. Vatsa, **Bethel Hall**, and William Eiers. “Neurosymbolic Characterization for Reliable Access Control Policy Analysis.” ISSRE, 2026.
3. **Bethel Hall et al.** “VeriFin: A Neurosymbolic Framework for Verifying LLM-Generated Financial Claims.” arXiv, 2026.
4. **Bethel Hall** and William Eiers. “Neurosymbolic Auditing of Natural-Language Software Requirements.” arXiv, 2026.

HONORS

- Selected as a *Global UGRAD Scholar* by the U.S. Department of State
- Palantir Global Impact Scholar, 2020

TALKS & PRESENTATIONS

Neurosymbolic Auditing of Natural-Language Software Requirements
AI for Mathematics and Computer Science Workshop (AIMACS)

July 2026
Lisbon, Portugal

CloudFix: Automated Policy Repair for Cloud Access Control Policies Using Large Language Models
SANER Conference 2026

March 2026
Limassol, Cyprus

PROFESSIONAL SERVICES

Teaching Experience:

- Teaching Assistant for Machine Learning
- Teaching Assistant for Java Programming

Spring 2025
Fall 2025

Conference Review:

- NeurIPS 2026
- ICAIF 2026